



COLLEGE
OF
MEDICINE

Measles and rubella

Assistant professor

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Objectives

□ To know about measles and rubella :

- Virus nature**
- Epidemiology**
- Symptoms and presentation**
- Identification of outbreak**
- Complication**

□ To be able to plan for :

- Control of any outbreak**
- Setting the preventive measures and its evaluation**

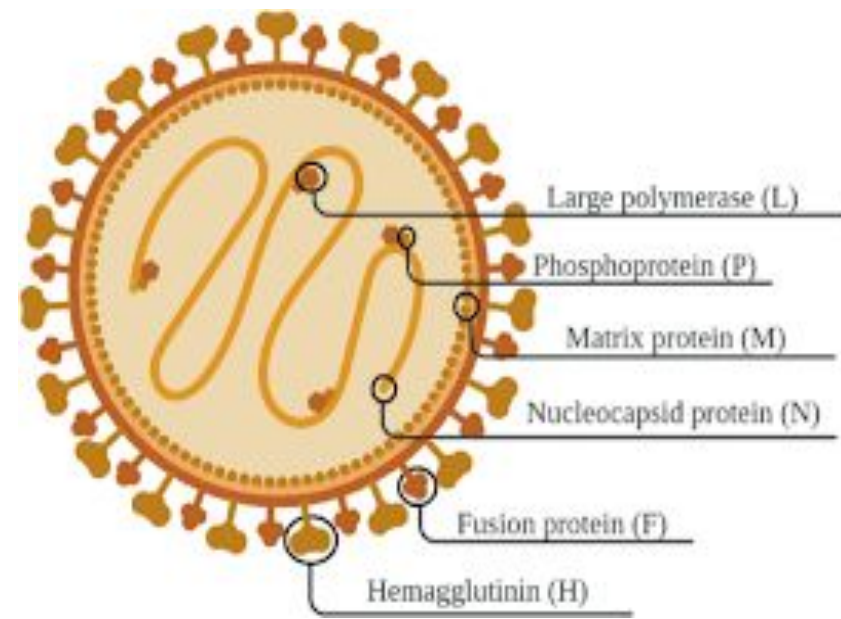
Measles

- It is highly contagious acute viral infection of respiratory tract caused by RNA virus Paramyxovirus (also known as rubeola which means red spots)
- Human is the only reservoir
- **Highly infectious** : susceptible person that's exposed to the measles virus has a 90% chance of becoming infected.
- **Highly contagious**: an infected person can go on to spread the virus to anywhere between 9 and 10 susceptible individuals (close contact)

historical Background

- **Rhazes in 10th century** who described in details the **symptoms and signs** of measles and smallpox based on clinical examination
- In 1954, **John F Enders and Dr Thomas C. Peebles** collected blood sample from several ill patients during measles outbreak in Boston, Massachusetts, they **succeed in isolating measles** in 13 years old boys blood (David Edmonston)

Measles Virus



- **Paramixovirus (genus *Morbillivirus*)**
- **Single stranded RNA virus.**
- Only one genotype.
- Rapidly inactivated by heat and light
- It has a short survival time (less than 2 hours) in the air or on objects and surfaces.

Epidemiology

- Occurrence : through out the world
- Reservoir: Human
- WHO (An estimated 136 000 people died from measles in 2022 – mostly unvaccinated or under vaccinated children of < five years) { out of 9 million cases}
- In 2023, there were over 322,714 cases of measles worldwide

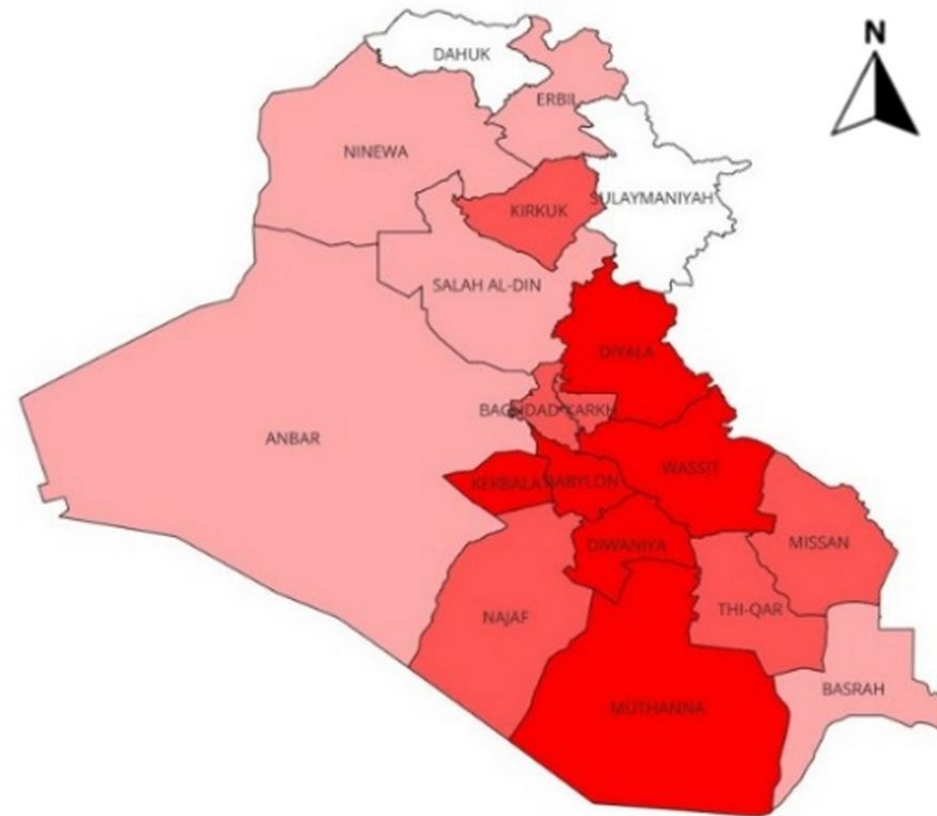
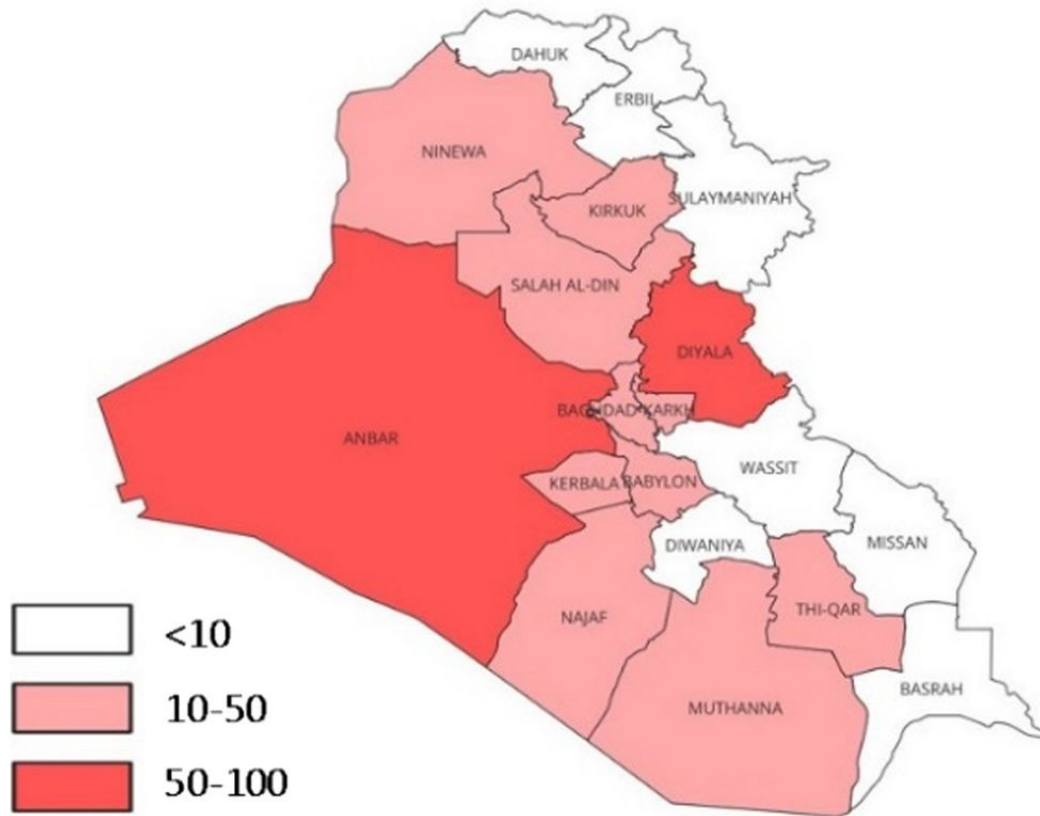
Measles cases in Iraq (2023- 2024)

- The incidence of measles in Iraq increased significantly, rising from (9579 cases) 22.1 per 100,000 population in 2023 to (30,798 cases) 69.3 in 2024
- In contrast, rubella incidence slightly decreased from (108 cases) 25/10,000,000 to (15 cases) 3 / 10,000,000 population,

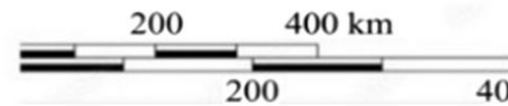
Measles in Iraq

2023

2024



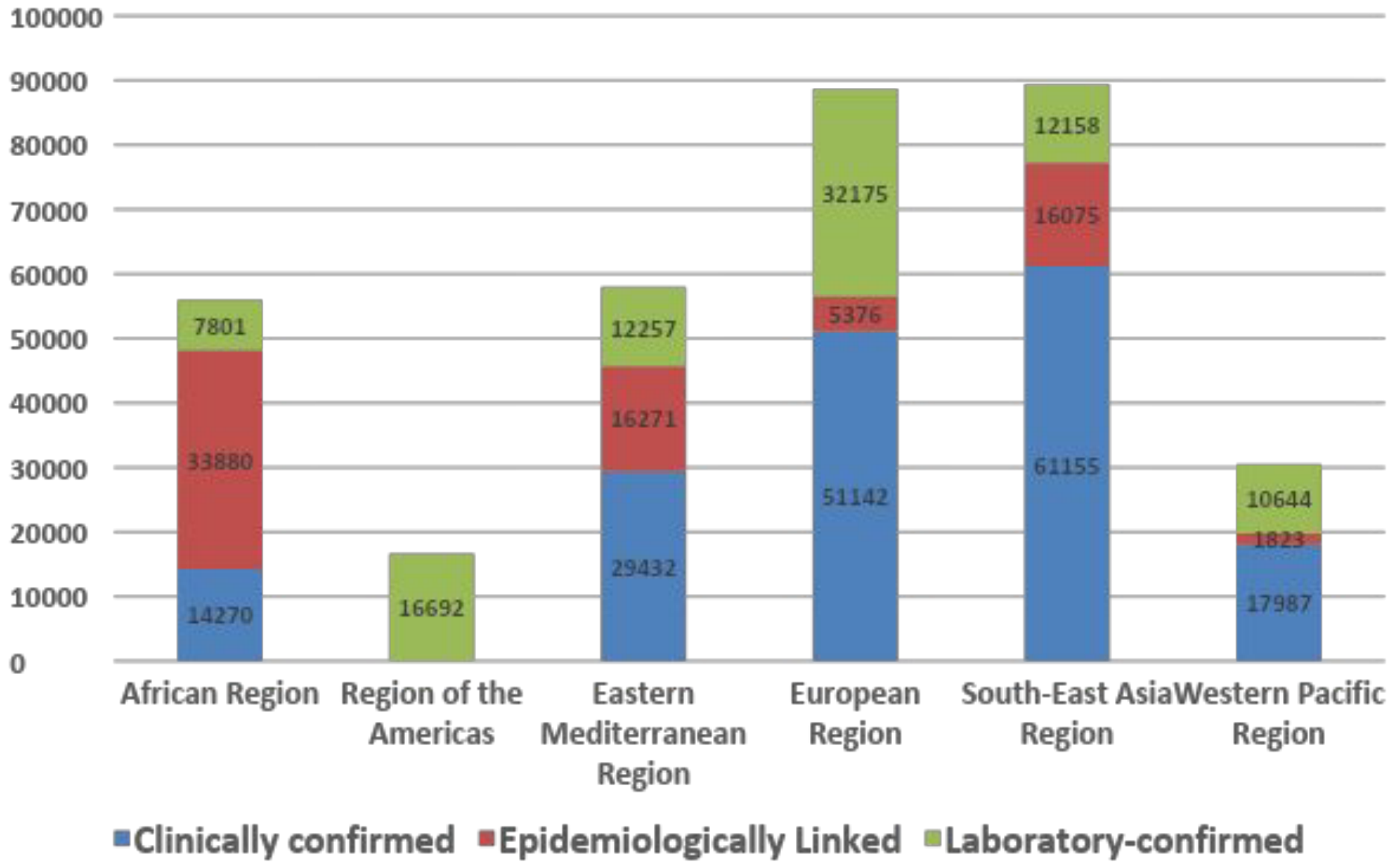
Iraq incidence rate of measles per 100,000 population (2023-2024)

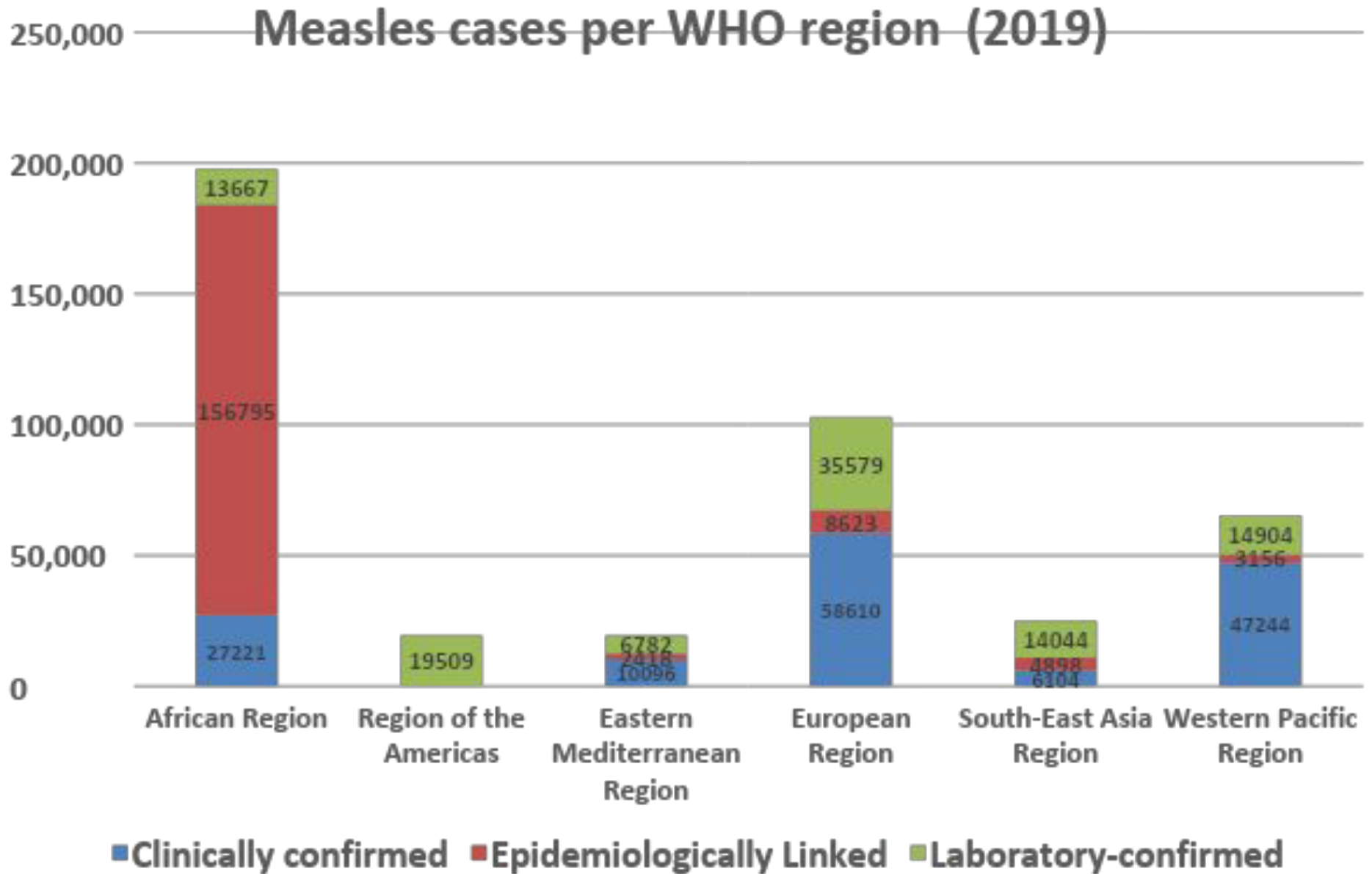


Epidemiology

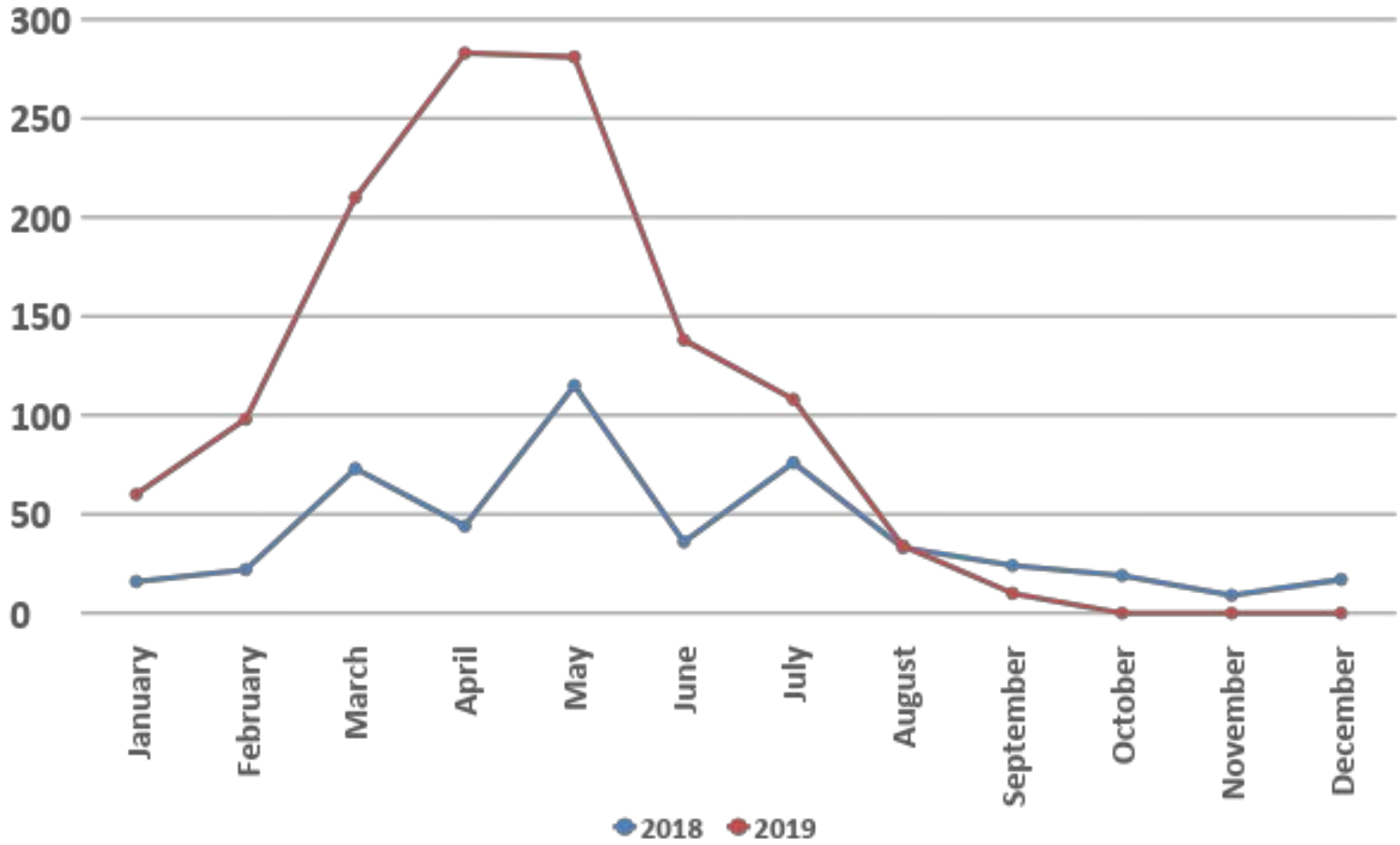
- Death rate ranges from (1- 2 death per 1000 cases in developed countries up to 30 – 60 deaths per 1000 measles cases in low income countries
- Transmission: **droplet borne**
- Virus-containing droplets can remain **in the air** for **several hours** and the virus remains infectious on **contaminated surfaces for up to two hours**
- Temporal pattern: More in Winter

Measles cases per WHO region (2018)





Measles cases in Iraq By months



Measles outbreak in Iraq

- Latest Measles outbreak in Iraq was during current year
- “Ministry of health recorded more than **27,000 cases of measles**, **96%** of which were among the unvaccinated, and 4% were among the vaccinated who did not complete the 2 doses.”
- In addition, he noted that **43 measles deaths** have been reported, of which 42 of the fatalities were unvaccinated.

Pathogenesis

- Measles is a systemic infection. The primary site of infection is the **respiratory epithelium of the nasopharynx**.
- 2-3 days after, a primary **viremia** occurs with subsequent infection of the reticuloendothelial system.
- a second viremia occurs 5–7 days after initial infection. During this viremia, there may be infection of the respiratory tract and other organs.

Communicability and incubation period

- Periods of communicability starts from the prodromal stage (4 – 5 days before appearance of rash) until 3–4 days after rash onset
- Incubation Period (10-12 Days)

Clinical Features

- Prodrome stage (2-4 Days).
 - Fever(39.4° C or more)
 - Cough, Coryza, Conjunctivitis → catarrhal inflammation of mucous in nose
- Koplik spot: if present it is pathognomonic,
(1-2 days before and 1-2 after appearance of rash
(blue –white spot on bright red background in buccal mucosa)
- Maculopapula Rash





Maculopapular Rash

- The rash is a maculopapular eruption that usually lasts 5–6 days
- It begins at the hairline, then involves the face and upper neck. During the next 3 days, the rash gradually proceeds downward and outward, reaching the hands and feet.
- The rash fades in the same order that it appears, from head to extremities.

Complications

- 30% of cases have at least one complication or more
- It is more in under 5 children and adult 20 years or older
- ❖ Diarrhea 8%
- ❖ Otitis media 7% (almost in children)
- ❖ Pneumonia 6%

Complications (continued)

- ❖ **Encephalitis 1/1000:** (6 D after the rash, characterized by fever, vomiting, stiff neck, Meningeal irritation, drowsiness, convulsion & coma) *CFR = 15%* case fatality rate
- ❖ **Seizure 0.1 %** (with or without fever)
- ❖ **Death 1 -3 / 1000** (malnourished up to 25%)
(Death is due to Pneumonia in children & Acute Encephalitis in Adult)

Complications (continued)

- ❖ **SSPE** { Subacute sclerosing panencephalitis}: (5/1,000,000) 7 years (1 month-27 years), it is degenerative CNS disease from persistent viral infection of brain.

Progressive deterioration of behavior and intellect, followed by ataxia, ^{→ loss of control of body movements} myoclonic seizure and death.

- ❖ **Atypical Measles Syndrome:** (fever, pneumonia, pleural effusions, and edema.)

Complications (Summary)

- ❖ Diarrhea
- ❖ Otitis media
- ❖ Pneumonia
- ❖ **Encephalitis**
- ❖ **Seizure**
- ❖ **Death**
- ❖ **SSPE** { Subacute sclerosing panencephalitis }
- ❖ **Atypical Measles Syndrome**



Measles Notification

- Measles cases require a coordinated and robust public health response
- Measles is nationally notifiable and cases should be reported immediately to the appropriate health department
- Health departments has to assist with testing when measles is suspected

High risk groups

People at high risk for **severe illness and complications** from measles include:

- Infants and children aged < 5 years
- Adults aged > 20 years
- Pregnant women
- People with compromised immune systems, such as from leukemia and HIV infection

Diagnosis

1- Clinical diagnosis

2- Laboratory method

□ **Isolation:** Urine, nasopharynx, throat swab or heparinized blood *→ blood that anti-heparin has been added to it to prevent clotting*

□ **Serologic :** either by positive serological test for measles IgM antibody or by significant raise in IgG (ELISA)

3- Epidemiological Linked

Prevention

The best way to protect against **measles** is to receive **measles vaccination** (safe and effective)

- First MCV 1963 (inactivated) no more used after 1967
- Live attenuated vaccine by (Edmonston-Enders strain) : less fever and Rash
- MCV (monovalent measles vaccine or measles only
- MMR (Mumps, Measles and Rubella)
- MMRV (Mumps, Measles, Rubella and varicella)

Continuation

- MCV given 9 Months (85%) *measles*
12 Months (95%) *MMR1*
18 Months (98%) *MMR2*
- Each child has to receive two doses of measles (MMR) from age of 12 months.
- Dose given before 12 M is not countable

Schedule for MCV

- Rout of Administration: S.C. at the arm (45°)
- The dose 0.5 cc
- First Dose : Completed 9 Month
- Second Dose: 12 Months (MMR1)
- Third Dose: 18 Months (MMR2)

Adverse - Event - following - Immunization

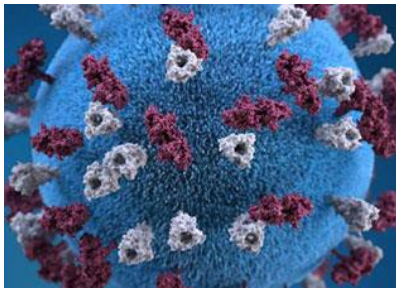
AEFI (MMR)

- 5-12 Days
- Fever (5%-15%) 39.4 ° c
- Rash (5%)
- Joint symptom (25%)
- Thrombocytopenia {15–35 days} (rare 1/30,000 doses)
- Parotitis (rare)
- Deafness (rare)
- Encephalopathy (less than one /1,000,000 doses)

Contra Indications

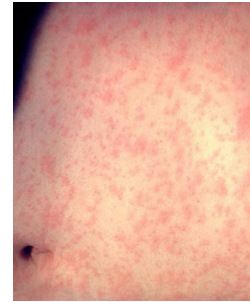
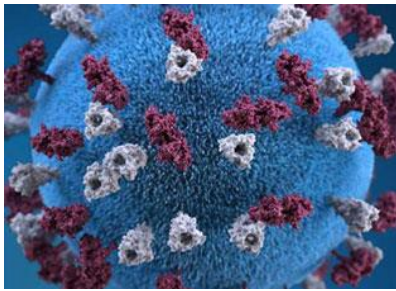
- ❑ Sever allergic reaction – vaccine component or prior dose
- ❑ Pregnancy
- ❑ Immune suppression
- ❑ Recently receiving blood product
- ❑ Moderate or sever acute illnesses

Measles – Typical Timeline



Incubation Period
10-14 days

Measles – Typical Timeline

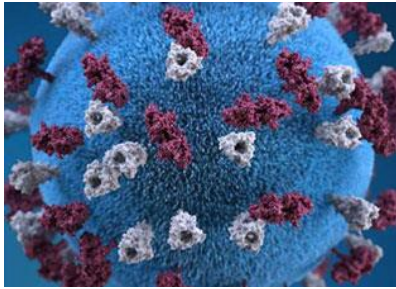


Day -4 to -2
Prodrome

Day 0

Measles – Typical Timeline

Contagious



Infectious Period:
4 days before rash to 4 days after rash

Other Common Causes of Maculopapular Febrile Rash

Human Herpesvirus 6 (Roseola)

Common cause of fever and rash in children; fever often resolves, and rash appears the next day starting on trunk and spreading outwards



Slapped Cheek rash

Enteroviruses

- Rash can be maculopapular or urticarial
- Often on hands/feet (Hand Foot Mouth, photo)



Antibiotic sensitivity or allergies



Rubella and Rubella Vaccine

Rubella

- From Latin meaning "little red"
- Discovered in 18th century - thought to be variant of measles
- First described as distinct clinical entity in German literature (German Measles)
- Congenital rubella syndrome described by Gregg in 1941

Rubella Virus

- Virus (*Togaviridae* family), Genus Rubivirus.
- RNA virus
- One antigenic type
- Rapidly inactivated by chemical agents, low pH, heat and ultraviolet light

Rubella Pathogenesis

- Respiratory transmission of virus
- Replication in nasopharynx and regional lymph nodes
- Viremia 5-7 days after exposure with spread to tissues
- Placenta and fetus infected during viremia

Rubella Epidemiology

- **Reservoir:** Human
- **Transmission:** Droplet borne or contact of respiratory secretions come in contact with hands which in turn come in contact to nasal mucosa
- **Temporal pattern:** Peak in late winter and spring
by winter like seasonal variation? could have other meanings elsewhere
- **Communicability:**
 - 7 days before to 5-7 days after rash onset.
 - Infants with CRS may shed virus for a year or more
by congenital Rubella Syndrome

Rubella Clinical Features

- Incubation period 14 days (range 12-23 days)
- Prodrome of low grade fever
- Lymphadenopathy in second week
- Maculopapular rash 14-17 days after exposure



Rubella Complications

- **Arthralgia or arthritis**
 - children rare
 - adult female up to 70 %
- **Thrombocytopenic purpura 1 / 3000 cases**
- **Encephalitis 1 / 5000 cases**
- **Neuritis rare**
- **Orchitis rare**

Congenital Rubella Syndrome

- Infection may affect all organs
- May lead to fetal death or premature delivery
- Severity of damage to fetus depends on gestational age
- Up to 85% of the fetuses were affected if infection occur during first trimester

Congenital Rubella Syndrome

- Deafness
- Cataracts
- Heart defects
- Microcephaly
- Mental retardation
- Bone alterations
- Liver and spleen damage

Rubella Laboratory Diagnosis

- Isolation of rubella virus from clinical specimen (e.g., nasopharynx, urine)
- Significant rise in rubella IgG by any standard serologic assay (e.g., enzyme immunoassay)
- Positive serologic test for rubella IgM antibody

Rubella Case Definition

- Acute onset of generalized maculopapular rash, and Temperature of >37.2 C , if measured
- and
- Arthritis/arthralgia or lymphadenopathy or conjunctivitis



- Laboratory diagnosis of rubella and CRS
- Step-by-step guidelines on evaluation and management of outbreak
- Rubella prevention and control among women of childbearing age
- Rubella and CRS surveillance

Prevention

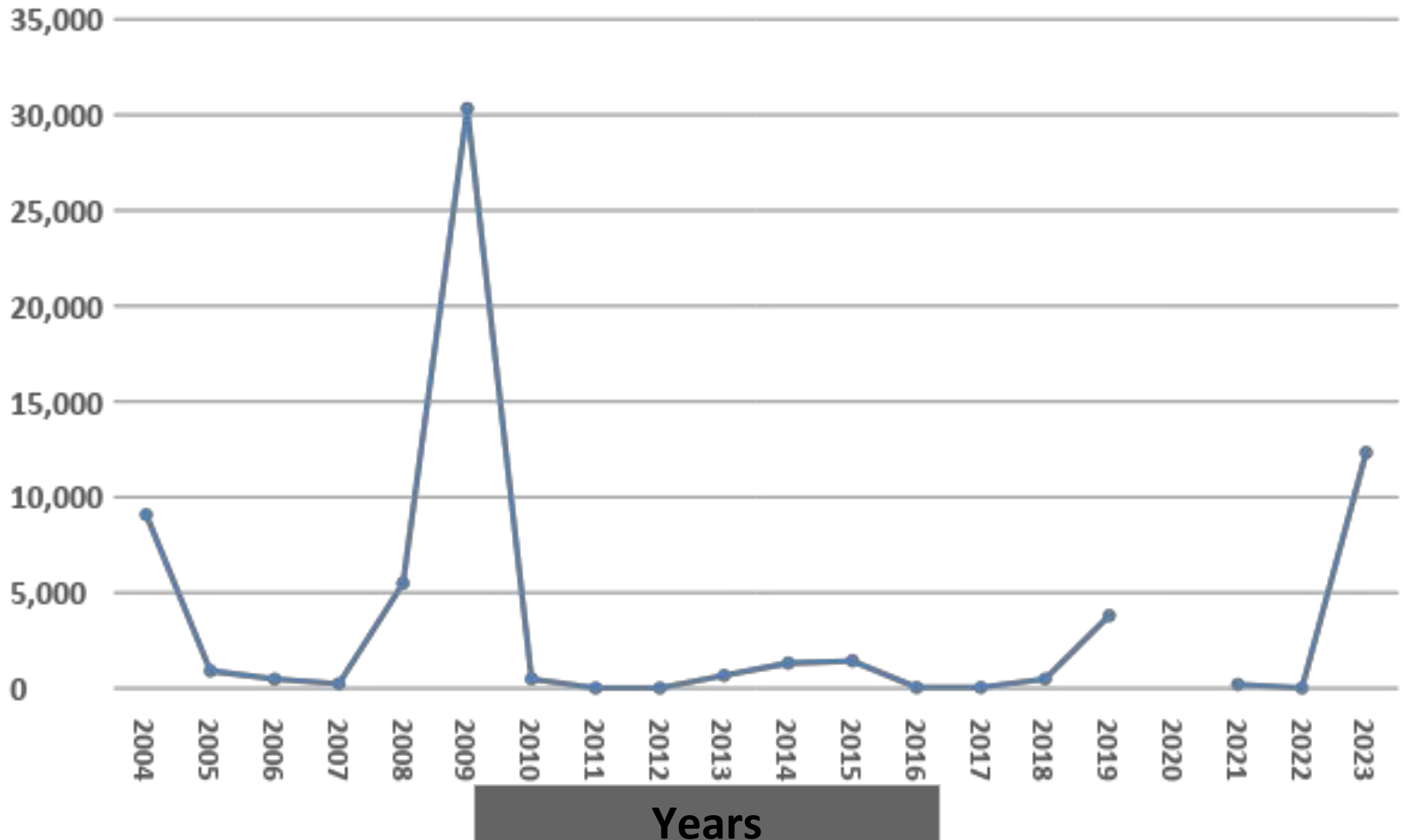
- Rubella Vaccine (Live attenuated vaccine) presented in MMR Vaccine.
- The dose is 0.5 CC given S.C. in the arm.
- The vaccine is given from 12 Month of age and onward.

Rubella Vaccine (MMR)

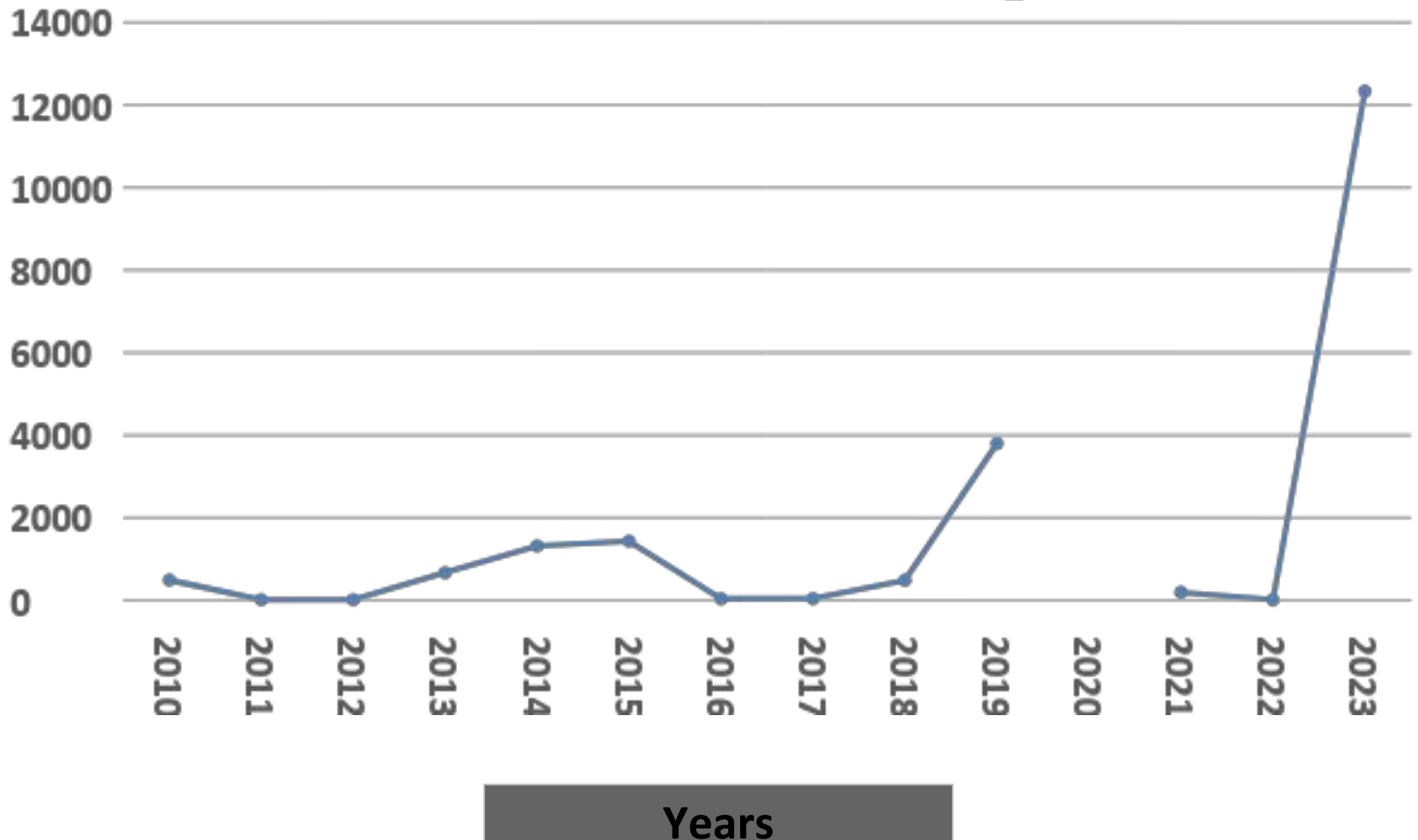
Indications

- All infants >12 months of age
- Susceptible adolescents and adults without documented evidence of rubella immunity
- Emphasis on non-pregnant women of childbearing age.

Measles cases in Iraq



Measles cases in Iraq



Measles cases in Iraq

